POC IDEA:

Use Case

In a customer-facing car rental platform available across web and mobile (Android & iOS), the release cadence has been accelerated from bi-weekly to weekly deployments.

The application maintains a regression suite of approximately 200 test cases, evenly distributed across Web, Android, and iOS platforms. With frequent feature integrations and code merges occurring in every release cycle, ensuring consistent quality without increasing execution overhead has become a critical requirement.

Problem Statement

With the shift to weekly releases, both feature validation and full regression testing are expected to be completed within significantly reduced timelines.

Executing the entire regression suite for every release is:

- Time-intensive
- Resource-heavy
- Potentially redundant for unaffected areas

This creates a bottleneck in maintaining release velocity while ensuring comprehensive quality coverage.

Proposed POC (Proof of Concept)

To address this challenge, the following approach is proposed:

- Extract and analyze all Pull Request (PR) details merged within each release cycle
- Identify impacted application areas based on code changes
- Develop an intelligent agent capable of:
  - Mapping PR changes to functional areas
  - Recommending a subset of regression test cases specifically relevant to the impacted modules

Value Proposition

- Enables **targeted regression testing** based on actual impact analysis
- Reduces execution effort on non-impacted test scenarios
- Improves **test efficiency without compromising coverage**
- Allows QA teams to focus on **high-risk and change-sensitive areas**
- Supports faster release cycles while maintaining product quality